

IC Engines Tutorial

1 Questions

1. In an Otto cycle air at 35°C and 0.1 MPa is compressed adiabatically. The compression ratio is 8. Heat added is 2100 kJ/kg . Calculate the air standard efficiency, the mean effective pressure for the cycle, the maximum temperature and pressure of the cycle. Assume $C_v = 0.717\text{ kJ/kgK}$ and $R = 8.314\text{ kJ/kmolK}$.
2. In an engine working on ideal Otto cycle the temperature at the beginning and end of compression are 50°C and 373°C . Find the compression ratio and the air standard efficiency of the engine. (Ans- $r = 5.66$, $\eta_{otto} = 50\%$.)
3. In an Otto cycle air at 17°C and 1 bar is compressed adiabatically until the pressure is 15 bar . Heat is added at constant volume, until the pressure rises to 40 bar . Calculate the air standard efficiency, the compression ratio and the mean effective pressure for the cycle. Assume $C_v = 0.717\text{ kJ/kgK}$ and $R = 8.314\text{ kJ/kmolK}$. (Ans. $\eta = 53.9\%$, $r = 6.91$, $p_m = 5.68\text{ bar}$)
4. Fuel supplied to an SI engine has a calorific value 42000 kJ/kg . The pressure in the cylinder at 30% and 70% of the compression stroke are 1.3 bar and 2.6 bar respectively. Assuming that the compression follows the law $pV^{1.3} = C$. Find the compression ratio. If the relative efficiency of the engine compared with the air standard efficiency is 50%. Calculate the fuel consumption in kg/kW.h . (Ans. $r = 4.68$ sfc= 0.373 kg/kWh).
5. A gas engine working on the Otto cycle has a cylinder of diameter 200 mm and stroke 250 mm . The clearance volume is 1570 cc . Find the air standard efficiency. Assume $C_v = 0.717\text{ kJ/kgK}$ and $C_p = 1.004\text{ kJ/kgK}$ for air. (Ans. $\eta = 51.2\%$)
6. In an SI engine working on the ideal Otto cycle, the compression ratio is 5.5. The pressure and temperature at the beginning of compression are 1 bar and 27°C respectively. The maximum pressure in the cycle is 30 bar . Determine the pressure and temperature at all the points, the air standard efficiency and the mean effective pressure. Assume ratio of specific heats to be 1.4 for air. (Ans. $p_2 = 10.88\text{ bar}$, $T_2 = 593.3\text{ K}$, $T_3 = 1635.7\text{ K}$, $p_4 = 1.98\text{ bar}$, $T_4 = 827\text{ K}$, $\eta_{otto} = 49.4\%$, $P_m = 5.25\text{ bar}$)

7. A gas engine operating on the ideal Otto cycle has compression ratio of 6:1. The pressure and temperature at the commencement of compression are 1 bar and 27 °C . The heat added during the constant volume combustion process is 1170 kJ/kg. Determine the maximum pressure and temperature of the cycle, work output per kg of air and air standard efficiency. Assume $C_v = 0.717$ kJ/kgK and $\gamma = 1.4$ kJ/kgK for air. (Ans. $T_3 = 2247$ K, $p_3 = 44.8$ bar, $W = 599$ kJ, $\eta_{otto} = 51.2\%$)
8. A spark ignition engine working on ideal Otto cycle has the compression ratio 6. The pressure and temperature at the beginning of compression are 1 bar and 37 °C respectively. The maximum pressure in the cycle is 30 bar. For unit mass flow, Calculate (i) p, V and T at all the points. (ii) the ratio of heat supplied to the heat rejected. Assume $R = 8.314$ kJ/kmolK and $\gamma = 1.4$ kJ/kgK for air. (Ans. $V_1 = 0.889$ m³, $p_2 = 12.3$ bar, $V_2 = 0.148$ m³, $T_2 = 634.8$ K, $p_3 = 30$ bar, $T_3 = 1548$ K, $p_4 = 2.44$ bar, $T_4 = 756.4$ K, $\frac{Q_s}{Q_r} = 2.04$)
9. In an Otto engine, the pressure and temperature at the beginning of compression are 1 bar and 37 °C respectively. Calculate the theoretical thermal efficiency of this cycle if the pressure at the end of the adiabatic compression is 15 bar. The maximum temperature of the cycle is 2000 K. Calculate (i) heat supplied per kg of air (ii) work done per kg of air and (iii) pressure at the end of adiabatic expansion. Assume $C_v = 0.717$ kJ/kgK and $\gamma = 1.4$ kJ/kgK for air. (Ans. $Q_s = 952.2$ kJ/kg, $w = 513$ kJ/kg, $p_4 = 2.98$ bar).
10. A diesel engine has a compression ratio of 20 and cut off takes place at 5% of the stroke. Find the air standard efficiency. Assume $\gamma = 1.4$ kJ/kgK for air. (Ans $\eta_{diesel} = 64.9\%$)
11. Determine the ideal efficiency of the diesel engine having a cylinder with bore 250 mm, stroke 375 mm and a clearance volume of 1500 cc, with fuel cut-off occurring at 5% of the stroke. Assume $\gamma = 1.4$ kJ/kgK for air. (Ans $\eta_{diesel} = 60.5\%$)
12. In an engine working on Diesel cycle inlet pressure and temperature are 1 bar and 17 °C respectively. pressure at the end of adiabatic compression is 35 bar. The ratio of expansion after constant pressure heat addition is 5. Calculate the heat addition , heat rejection and the efficiency of the cycle. Assume $\gamma = 1.4$ kJ/kgK $C_v = 0.717$ kJ/kgK and $C_p = 1.004$ kJ/kgK for air. (Ans. $Q_s = 1235.5$ kJ/kg, $Q_r = 557.5$ kJ/kg, $\eta_{diesel} = 54.9\%$)